Adopted Levels

Type	Author	History Citation	Literature Cutoff Date
Full Evaluation	Balraj Singh	NDS 156, 70 (2019)	31-Jan-2019

S(n)=7300 *SY*; S(p)=3250 *SY*; Q(α)=10370 *30* 2017Wa10
Estimated uncertainties (2017Wa10): ΔS(n)=760, ΔS(p)=1040.
S(2n)=13290 *930*, S(2p)=5350 *1040* (syst,2017Wa10).
2012Og06, 2006Og05, 2004Og10 (also 2011Og07, 2009Og07, 2007Og05, 2005Og03, 2005OgZZ, 2004OgZZ, 2004Mo15, 2003OgZZ, 2002OgZX): ^{286}Fl produced as α grand-daughter of ^{294}Og in four correlated decay chains. See ^{294}Og Adopted Levels for details.

2006Og05, 2004Og07, 2004Og05: ^{286}Fl produced as α daughter of ^{290}Lv in 12 correlated decay chains. See ^{290}Lv Adopted Levels for details.

2004Og12, 2004OgZZ, 2004Og10: ^{286}Fl produced in ^{242}Pu(^{48}Ca,4n),E=244,250 MeV reaction at FLNR-JINR-Dubna using DGFRS separator. A total of nine EVR-α/SF-SF correlated decay chains reported, seven at E=244 MeV and two at E=250 MeV. The α/SF event was attributed to ^{286}Fl in five of the decay chains.
History of EVR-α/SF-SF decay chain (2004Og12,2004OgZZ,2004Og10):

Event #1: Energy of the evaporation residue (EVR)=10.9 MeV (2004Og12).
$E_{\alpha 1}$=10.10 MeV *15*, $\Delta t_{\alpha 1}$=14.2 ms, assigned to ^{286}Fl α decay.
E_{SF}=213 MeV, Δt_{SF}=0.978 ms, assigned to ^{282}Cn SF decay.

Event #2: Energy of the evaporation residue (EVR)=11.5 MeV (2004Og12).
E_{SF}=201 MeV, Δt_{SF}=181 ms, assigned to ^{286}Fl SF decay.

Event #3: Energy of the evaporation residue (EVR)=13.0 MeV (2004Og12).
$E_{\alpha 1}$=10.25 MeV *10*, $\Delta t_{\alpha 1}$=84.7 ms, assigned to ^{286}Fl α decay.
E_{SF}=189 MeV, Δt_{SF}=0.458 ms, assigned to ^{282}Cn SF decay.

Event #4: Energy of the evaporation residue (EVR)=12.9 MeV (2004Og12).
E_{SF}=162 MeV, Δt_{SF}=135 ms, assigned to ^{286}Fl SF decay.

Event #5: Energy of the evaporation residue (EVR)=10.3 MeV (2004Og12).
$E_{\alpha 1}$=10.20 MeV *6*, $\Delta t_{\alpha 1}$=185 ms, assigned to ^{286}Fl α decay.
E_{SF}=164 MeV, Δt_{SF}=1.586 ms, assigned to ^{282}Cn SF decay.

Event #6: Energy of the evaporation residue (EVR)=13.3 MeV (2004Og12).
E_{SF}=218 MeV, Δt_{SF}=41.4 ms, assigned to ^{286}Fl SF decay.

Event #7: Energy of the evaporation residue (EVR)=8.9 MeV (2004Og12).
E_{SF}=182 MeV, Δt_{SF}=335 ms, assigned to ^{286}Fl SF decay.

Event #8: Energy of the evaporation residue (EVR)=10.5 MeV (2004Og12).
E_{SF}=228 MeV, Δt_{SF}=603 ms, assigned to ^{286}Fl SF decay.

Event #9: Energy of the evaporation residue (EVR)=13.3 MeV (2004Og12).
$E_{\alpha 1}$=10.34 MeV *22*, $\Delta t_{\alpha 1}$=150 ms, assigned to ^{286}Fl α decay.
E_{SF}=211 MeV, Δt_{SF}=0.143 ms, assigned to ^{282}Cn SF decay.

Adopted Levels (continued)

2010El06, 2009St21: ^{286}Fl produced in ^{242}Pu(^{48}Ca,4n), E=256, 263 MeV at LBNL cyclotron facility using the BGS separator. Signals from multiwire proportional counters (MWPC) and focal plane detector (FPD) were used to distinguish implantation events from radioactive decay events in the FPD. Z=114 events were identified by detecting time and position correlated events corresponding to their implantation and subsequent radioactive decay chain terminating in SF decay.

History of decay chains in 2010El06 and 2009St21:

Event #1: Energy of the evaporation residue (EVR)=14.37 MeV (2010El06).
$E_{\alpha 1}$=10.31 MeV *10*, Δt_1=76 ms, assigned to ^{286}Fl α decay.
E(SF)=205.4 MeV, Δt(SF)=522 μs, assigned to ^{282}Cn SF decay.

Event #2: Energy of the evaporation residue (EVR)=11.55 MeV (2009St21).
$E_{\alpha 1}$=10.23 MeV *4*, Δt_1=0.3009 s, assigned to ^{286}Fl α decay.
E(SF)=214.5 MeV, Δt(SF)=3.6 ms, assigned to ^{282}Cn SF decay.

For theoretical studies, consult Nuclear Science References (NSR) database at NNDC, BNL for 154 primary references dealing with the half-lives and other aspects of nuclear structure in this mass region.

^{286}Fl Levels

Cross Reference (XREF) Flags

A ^{290}Lv α decay (8.3 ms)

E(level)	J^π	$T_{1/2}$	XREF	Comments
0	0^+	166 ms *+40−27*	A	%α=60 *+10−11*; %SF=40 *+11−10* (2017Og01,2015Og05) Decay modes: others: %α=52, %SF=48 (2016Ho09 review); %α=50, %SF=50 (2012Og06). E(level): the reported activity is assumed to belong to the g.s. of ^{286}Fl. $T_{1/2}$: from 2016Ho09 review, analysis of all the 27 correlated decay chains reported, 25 in JINR-Dubna work and two in LBNL-Berkeley work. Other: 0.12 s *+4−2* (2017Og01 and 2015Og05 reviews, also 2012Og06 from analysis of 20 chains out of a total of 25 observed correlated decay chains in 2012Og06, 2006Og05, 2004Og07 and 2004Og12). Eα=10.21 MeV *4* (2017Og01, 2015Og05 and 2016Ho09 reviews). Measurements: 10190 *60* (2012Og06, from analysis of 12 events out of a total of 25 observed events in Dubna work, one in 2012Og06 study, and the rest in previous studies at Dubna); 10.21 MeV *6* (2004Og12); 10.23 MeV *4* (2009St21); 10.31 MeV *10* (2010El06). Production σ=1.4 pb *+32−12* in ^{242}Pu(^{48}Ca,4n),E=244 MeV at center-of-target (2009St21), later adjusted to 3.1 pb *+49−26* by 2010El06; 0.6 ps *+9−5* in ^{242}Pu(^{48}Ca,4n),E=256 MeV (2010El06); 4.5 pb *+36−19* in ^{242}Pu(^{48}Ca,4n),E=238 MeV (2004Og12).